

HIRAM ZUNIGA

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RESEARCH INTERESTS

Computer vision, with applications in medical imaging and assistive robotics. Vision-language models and retrieval-augmented multimodal systems; generative models and diffusion; quantum machine learning.

EDUCATION

CETYS University

2022 – 2026

B.S. in Computer Science Engineering

Tijuana, Mexico

- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Evolutionary Computation, Parallel Computing, Analysis of Algorithms, Probability & Statistics, Data Structures, Databases.

Yonsei University – Winter Abroad Program

Dec 2023 – Jan 2024

Big Data Analysis

Seoul, South Korea

PUBLICATIONS & MANUSCRIPTS

First-author works marked with *. Replace “et al.” with full author lists.

*[1] **H. Zuniga**, et al. “A Comparison of CNN Architectures for Alzheimer’s Disease Detection in Single-View MRI Scans.” *SPIE Optics + Photonics 2026, Applications of Machine Learning*, San Diego, CA. **Accepted – oral presentation.**

[2] [Author list], incl. **H. Zuniga**. “Retrieval-Augmented Vision-Language Models for Robust Leukemia Cell Classification Across Multiple Microscopy Datasets.” *SPIE Optics + Photonics 2026, Applications of Machine Learning*, San Diego, CA. **Accepted – oral presentation.**

*[3] **H. Zuniga**, et al. “A Distributed Human-Following Robot Using Deep Learning and Sensor Fusion.” *KES 2026, Procedia Computer Science (CORE B)*. **Accepted – camera-ready.**

*[4] **H. Zuniga**, et al. “IoT-Based System Using Quantum Support Vector Machine for Type-2 Diabetes Classification.” *Advances in Computers* (Elsevier, Q1 book series), Chapter 10. **Revised manuscript submitted (major revision addressed).**

[5] [Author list], incl. **H. Zuniga**. “A Human-Following Robot for Indoor Navigation Using Deep Learning and LiDAR-Based Path Planning.” *SPIE Security + Defence 2026, Autonomous Systems for Security and Defence*. **Under review.**

RESEARCH EXPERIENCE

Research Assistant – Computer Vision

Oct 2025 – Present

CETYS University

Tijuana, Mexico

- Fine-tuned a vision-language model and designed a retrieval-augmented generation (RAG) pipeline for leukemia cell classification, evaluating robustness across multiple microscopy datasets [2].
- Benchmarked classical CNN architectures for Alzheimer’s disease detection in single-view MRI scans [1].
- Leading development of a socially assistive human-following robot for the elderly: deep-learning perception, sensor fusion, CNN-based identity verification, and an LLM conversational interface [3], [5].
- Implemented and evaluated a Quantum Support Vector Machine for Type-2 diabetes classification within an IoT acquisition system [4].

Machine Learning Capstone – Augmented AI for Thematic Analysis

Aug 2025 – Present

California State University, Los Angeles

Los Angeles, CA (remote)

- Built an end-to-end NLP pipeline: SentenceTransformers embeddings, UMAP dimensionality reduction, clustering, and Optuna hyperparameter optimization, producing theme maps from raw interview/review corpora.
- Designed LLM-assisted labeling and summarization with controlled granularity via a hierarchical label tree.

HONORS & AWARDS

3rd place, MIT iQuHACK 2026, Quantum Rings track – Massachusetts Institute of Technology, Jan–Feb 2026. ML models predicting quantum-algorithm runtime and minimum fidelity thresholds from QASM circuit features.

3rd place, Yale YQuantum 2026 – Yale University, Apr 2026. QAOA for portfolio optimization.

Regional 1st place & Global Nominee, NASA International Space Apps Challenge – Oct 2025. *ExoVision*: end-to-end ML toolkit for exoplanet detection (6-classifier comparison, public Streamlit app).

2nd place, GEAR Web3 Regional Hackathon – Ensenada, Mexico, Jun 2023 (USD \$1,000 prize).

Selected, Enactus National Hackathon 2026 (via Swift Changemakers Hackathon) – *BIMBO en Camino*, an intelligent distribution assistant (iOS, LLMs, speech interfaces).

NVIDIA Certificate: *Getting Started with Deep Learning*.

TALKS & PRESENTATIONS

Oral presentations (×2), *SPIE Optics + Photonics 2026*, San Diego, CA – upcoming.

Technical demo, Amazon San Diego office – presented *Message Router*, an event-driven AWS pipeline (EventBridge, SNS, SQS, Lambda, Step Functions), to AWS engineers and mentors, 2025.

SELECTED PROJECTS

Navi – assistive-robotics prototype for blind and visually impaired users: YOLO-based perception, LiDAR, VLM scene description, and LLM voice interaction; designed the real-time inference procedure. HackMerced XI, UC Merced, Mar 2026.

ExoVision – interactive ML toolkit for exoplanet detection on NASA open data; public app: exo-vision.streamlit.app.

OpenHelp – public-health platform for vulnerable communities (shelter maps, weather alerts, QR health summaries). Big Data Hackathon for San Diego, SDSU, Nov 2025.

Additional ML projects (RAG study assistant, tuberculosis CNN classifier, semantic book explorer) at github.com/HiramZ04.

INDUSTRY EXPERIENCE

Data Analyst	Mar 2025 – Sep 2025
<i>I3PL (Cross-Border Logistics Solutions)</i>	<i>Tijuana, Mexico</i>
Automated multi-API reporting pipelines in Python (80% reduction in manual work); built company-wide dashboards and a validated data-capture system replacing spreadsheets.	

TECHNICAL SKILLS & LANGUAGES

Programming: Python, SQL, Swift. **ML/CV:** PyTorch, scikit-learn, OpenCV, YOLO, VLMs, CNNs, RAG, Hugging Face Transformers, SentenceTransformers, UMAP, Optuna, Ollama. **Cloud:** AWS (SageMaker, Bedrock, Lambda, Step Functions, EventBridge, SQS, SNS, S3, DynamoDB, RDS). **Quantum:** Qiskit, QAOA, QSVM.

Languages: Spanish (native), English (TOEFL iBT 100/120).